

2024 Paper 2 Questions by Topic

- 3 Lightning occurs when charge builds up in the atmosphere, creating a potential difference between the ground and the atmosphere.

During a lightning strike there is an average current of $3.3 \times 10^4 \text{ A}$ for a time of $2.6 \times 10^{-5} \text{ s}$.

- (a) Calculate the charge transferred during the lightning strike.

3(a)	$Q = It$	C1
	$= 3.3 \times 10^4 \times 2.6 \times 10^{-5}$	A1
	$= 0.86 \text{ C}$	

charge = C [2]

- (b) The potential difference between the ground and the atmosphere is $3.0 \times 10^7 \text{ V}$.

Calculate the average power, in GW, transferred during the lightning strike.

3(b)	$P = IV$ or $P = VQ / t$ or $V = IR$ and $P = V^2 / R$ or $P = I^2 R$	C1
	$P = 3.3 \times 10^4 \times 3.0 \times 10^7$ or $P = (3.0 \times 10^7 \times 0.86) / (2.6 \times 10^{-5})$ or $P = (3.0 \times 10^7)^2 / 910$ or $P = (3.3 \times 10^4)^2 \times 910$ $P = 9.9 \times 10^{11} \text{ (W)}$ $= 990 \text{ GW}$	A1

..... GW [2]

- (c) A lightning rod is attached to a tall building to conduct charge safely to the ground. The lightning rod is modelled as a uniform cylindrical copper cable of total length 95m that runs from the ground to the top of the building, as shown in Fig. 3.1.

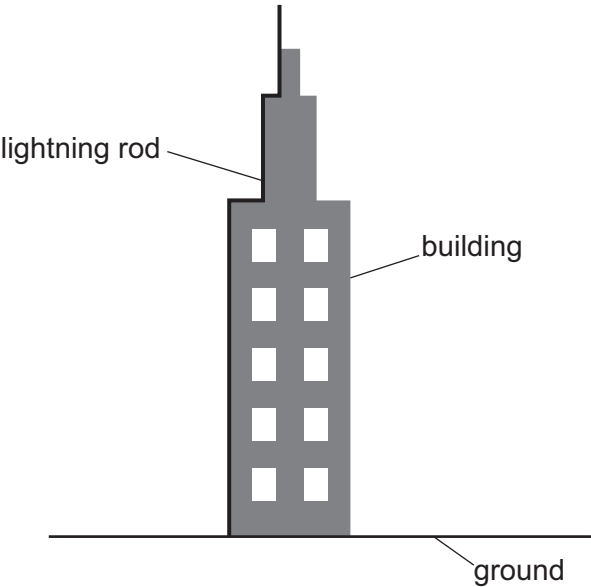


Fig. 3.1

- (i) The resistance of the lightning rod is $9.6\,\Omega$.
The resistivity of copper is $1.7 \times 10^{-8}\,\Omega\text{m}$.

Determine the radius of the lightning rod.

3(c)(i)	$R = \rho L / A$	C1
	$9.6 = 1.7 \times 10^{-8} \times 95 / \pi r^2$	C1
	$r = 2.3 \times 10^{-4}\text{ m}$	A1

radius = m [3]

- (ii) The radius of the copper lightning rod is doubled with no change to its length.

State the effect of this change on the resistance of the lightning rod.

3(c)(ii)	(resistance) decreases by a factor of four	A1
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 [1]

- (d) A section of the lightning rod of length 0.12m is removed for testing. A tensile stress of $1.9 \times 10^6\text{Pa}$ is applied, as shown in Fig. 3.2.

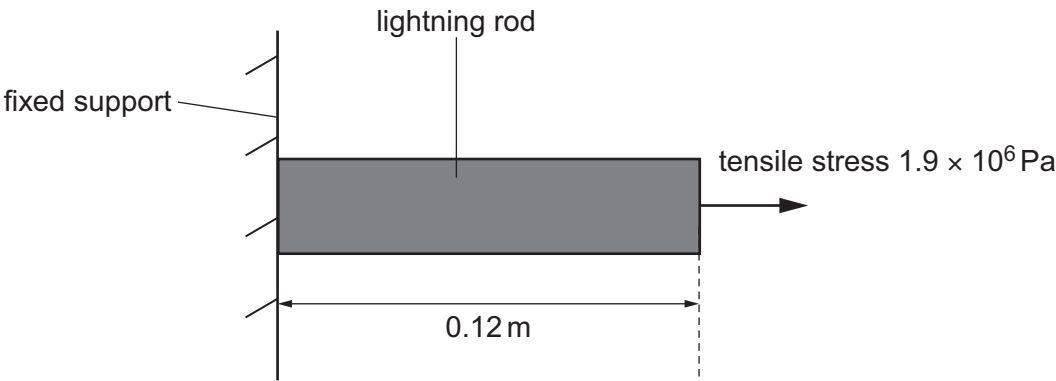


Fig. 3.2 (not to scale)

The section of the rod obeys Hooke's law. The Young modulus of copper is $1.3 \times 10^{11}\text{Pa}$.

Calculate the extension of the section.

3(d)	$E = \sigma / \epsilon$	C1
	$x = \sigma L / E$	C1
	$= 1.9 \times 10^6 \times 0.12 / (1.3 \times 10^{11})$	
	$= 1.8 \times 10^{-6}\text{m}$	A1

extension = m [3]

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6 (a) Define resistance.

potential difference per unit current	B1
..... [1]	

(b) A cylindrical metal wire of length 2.4 m and cross-sectional area $8.0 \times 10^{-6} \text{ m}^2$ has a resistance of 0.33Ω . There is a current in the wire of 4.7 A.

(i) Determine the resistivity of the metal from which the wire is made.

$\rho = RA / L$	C1
$= (0.33 \times 8.0 \times 10^{-6}) / 2.4$	A1
$= 1.1 \times 10^{-6} \Omega \text{ m}$	

resistivity = $\Omega \text{ m}$ [2]

(ii) Calculate the charge that passes through the wire in a time of 5.0 minutes.

$Q = It$	C1
$= 4.7 \times 5.0 \times 60$	A1
$= 1400 \text{ C}$	

charge = C [2]

(iii) The free electrons (charge carriers) in the wire have an average drift speed of 0.16 mm s^{-1} .

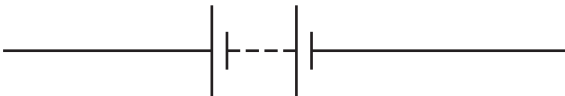
Determine the number density of charge carriers in the metal.

$I = nAvq$	C1
$n = 4.7 / (8.0 \times 10^{-6} \times 0.16 \times 10^{-3} \times 1.60 \times 10^{-19})$	A1
$= 2.3 \times 10^{28} \text{ m}^{-3}$	

number density = m^{-3} [2]

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- (c) The wire in (b) may be considered to be a fixed resistor. It is connected in series with a thermistor to a battery that has negligible internal resistance.
- (i) Use circuit symbols to complete Fig. 6.1 to show the circuit diagram of this arrangement.



correct symbols for resistor and thermistor, shown correctly connected in series with the battery	B1
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Fig. 6.1

[1]

- (ii) Explain, without calculation, how the power dissipated in the wire changes as the temperature of the thermistor is increased.

(as temperature increases) resistance of thermistor decreases	M1
(total resistance decreases so) greater current (in circuit/wire) so power (dissipated in the wire) increases or (total resistance decreases so) greater (share of) p.d. across wire so power (dissipated in the wire) increases	A1

..... [2]

[Total: 10]

7 (a) Define electric potential difference.

energy (transferred) per (unit) charge

B1

.. [1]

(b) A cell of electromotive force (e.m.f.) 1.8V and internal resistance r is connected in parallel with a resistor of resistance 6.0Ω and a filament lamp, as shown in Fig. 7.1.

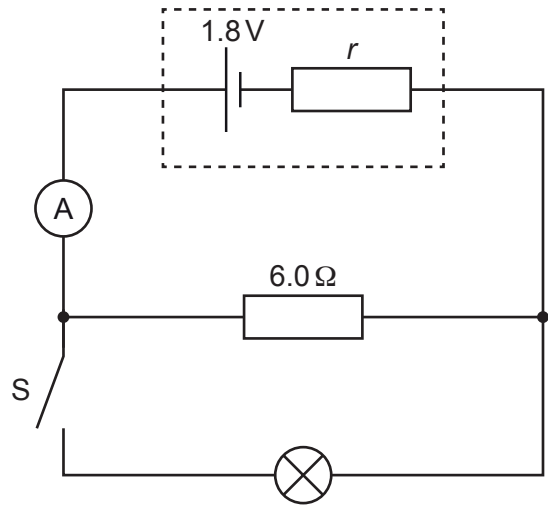


Fig. 7.1

The switch S is open. The ammeter reading is 0.25A.

Determine the internal resistance r of the cell.

$V = 0.25 \times 6$ $= 1.5$ $Ir = E - IR$ $Ir = 1.8 - 1.5$ $= 0.3$ $r = 0.3 / 0.25$ $= 1.2\Omega$	C1
OR	
(Total) $R = 1.8 / 0.25$ $= 7.2$ $E/I = (R + r)$ $1.8/0.25 = 6 + r$ $r = 7.2 - 6$ $= 1.2\Omega$	(C1)
	(C1)
	(A1)

$r = \dots\dots\dots \Omega$ [3]

(c) At time t_1 switch S in Fig. 7.1 is closed. Fig. 7.2 shows the variation with time t of the ammeter reading I .

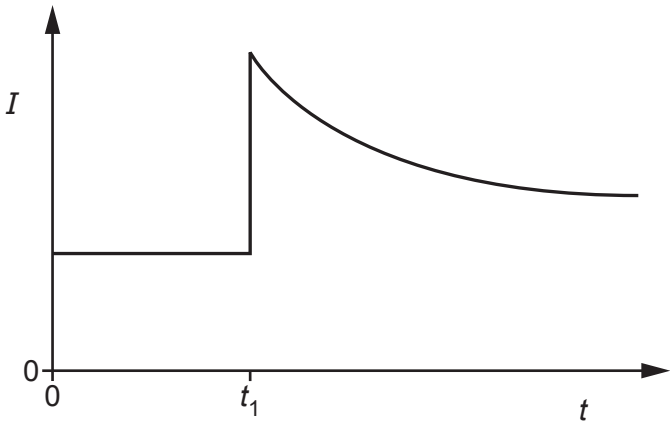


Fig. 7.2

(i) State whether the e.m.f. of the cell after t_1 is greater than, less than or the same as it was before t_1 .

The same	B1 [1]
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(ii) By considering the effect of the lamp on the total resistance of the circuit, explain the variation of the ammeter reading shown in Fig. 7.2.

- Any 3 points from:
- before t_1 / when current constant, the (total) resistance is constant
 - at t_1 / when current increases, the (total) resistance decreases (due to decrease of external resistance)
 - (after t_1) temperature (of lamp) increases (so the resistance of the lamp increases) [3]
 - (after t_1) resistance of lamp increases (so total resistance increases so the current in the ammeter decreases) [Total: 8]

6 (a) State Kirchhoff's first law.

sum of current(s) entering a junction = sum of current(s) leaving (the same junction)
or
(algebraic) sum of current (s) at a junction is zero

B1 ...

[1]

(b) A cell with internal resistance r is connected to two resistors of resistances R_1 and R_2 as shown in Fig. 6.1.

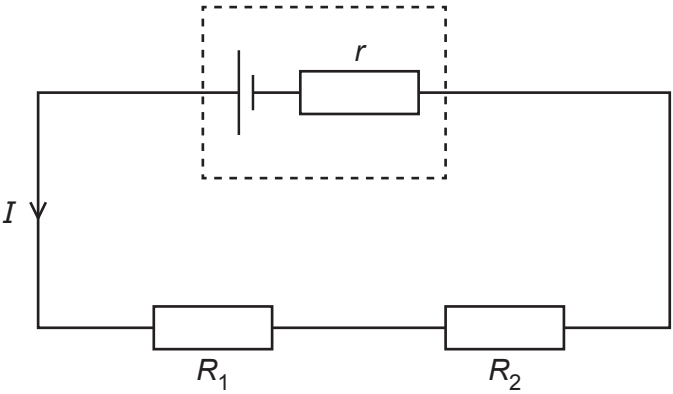


Fig. 6.1

The potential differences (p.d.s) across R_1 and R_2 are V_1 and V_2 respectively.
The terminal p.d. across the cell is V .
The current in the circuit is I .

Use Kirchhoff's laws to show that the total resistance R_T of the external circuit is given by

$$R_T = R_1 + R_2 .$$

(by Kirchhoff's second law) $V = V_1 + V_2$

B1

so $IR_T = IR_1 + IR_2$ (and cancelling I gives) $R_T = R_1 + R_2$

B1

or

$V/I = V_1/I + V_2/I$ (and substituting R gives) $R_T = R_1 + R_2$

[2]

(c) The electromotive force (e.m.f.) of the cell in Fig. 6.1 is 1.50 V.

The values of R_1 and R_2 are $10\,\Omega$ and $15\,\Omega$ respectively. The terminal p.d. of the cell is 1.35 V.

Calculate the internal resistance r of the cell.

current in circuit = $1.35 / (10 + 15)$ (= 0.054 A)	C1
$r = (E - V) / I$	C1
= $(1.5 - 1.35) / 0.054$	A1
= $2.8\,\Omega$	
or	
by potential divider principle $\frac{0.15}{1.35} = \frac{r}{25}$	(C2)
$r = 2.8\,\Omega$	(A1)
or	
$I = 1.35 / (10 + 15)$ (= 0.054 A)	(C1)
total resistance = $1.50 / 0.054$ (= 27.8 Ω)	(C1)
$r = 27.8 - 25$	
$r = 2.8\,\Omega$	(A1)

..... Ω [3]

(d) A resistor of resistance R_3 is added to the circuit in Fig. 6.1, so that the circuit is as shown in Fig. 6.2.

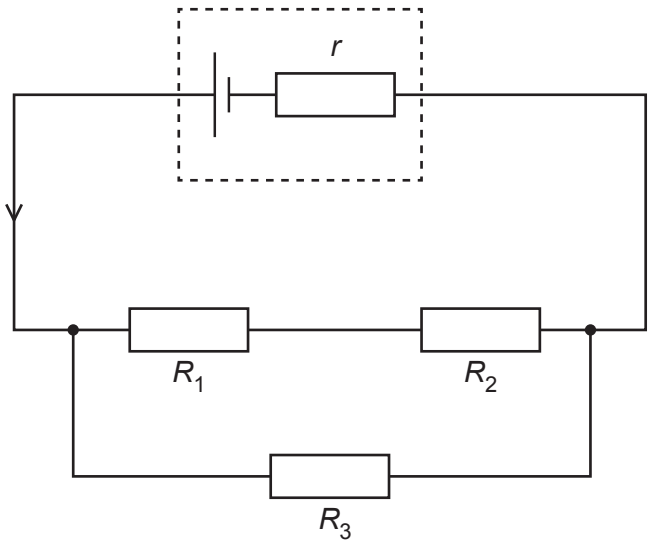


Fig. 6.2

State and explain the effect, if any, of this change on:

(i) the current in the cell

the (total) resistance (of the circuit) has decreased (and e.m.f. is unchanged)	M1
(the current (in the cell) will) increase	A1

..... [2]

(ii) the terminal p.d. of the cell.

(as the current is greater and so there is a) larger p.d. across the internal resistance	M1
(terminal p.d. will) decrease	A1

..... [2]

- 5 (a) State Kirchhoff's second law.

sum of electromotive force(s) = sum of potential difference(s) around a (closed) loop.

B1

[1]

- (b) A battery of electromotive force (e.m.f.) 9.0V and negligible internal resistance is connected in series with a variable resistor X and a thermistor Y as shown in Fig. 5.1.

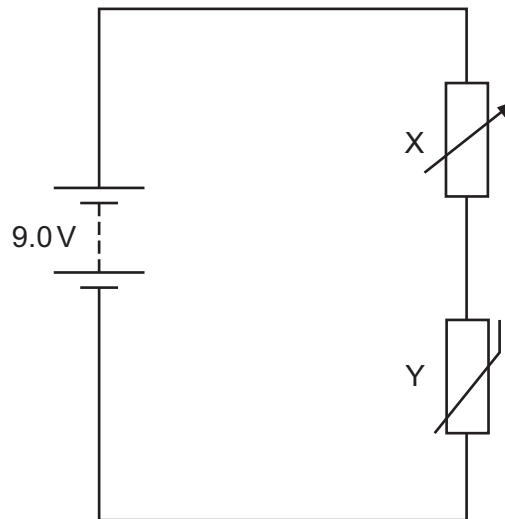


Fig. 5.1

Fig. 5.2 shows the relationship between temperature and resistance for the thermistor.

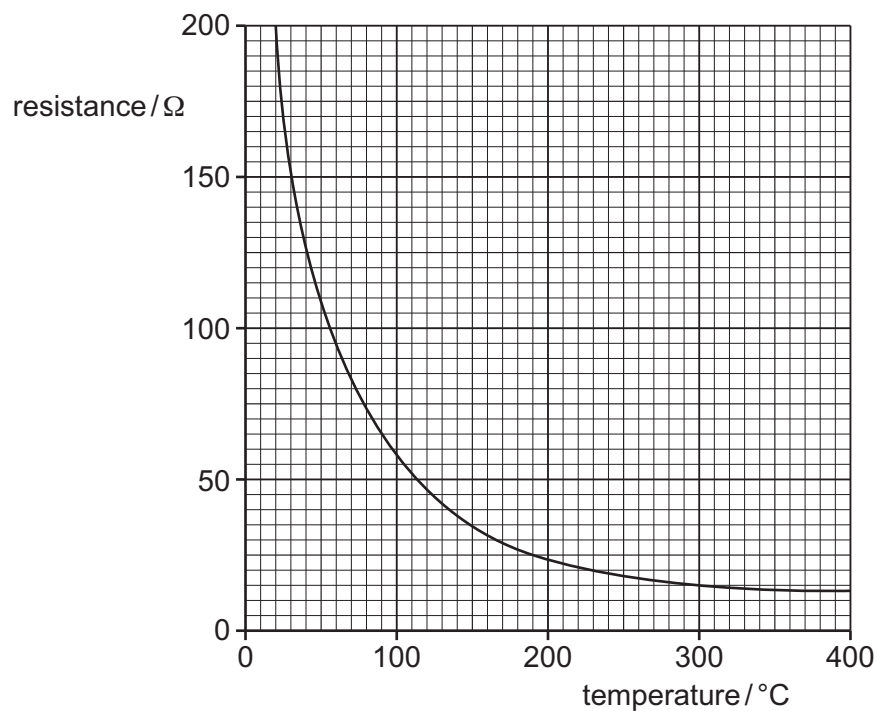


Fig. 5.2

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- (i) The current in the circuit is $1.1 \times 10^{-2}\text{A}$. The potential difference across Y is 4.0V .
Calculate the resistance of X.

$R = V / I$	C1
$= (9.0 - 4.0) / 1.1 \times 10^{-2}$	A1
$= 450 \Omega$	

resistance = Ω [2]

- (ii) The temperature of Y is changed to 190°C . The resistance of X remains unchanged.

Determine the new potential difference across Y.

resistance (of thermistor) = $25 (\Omega)$ (from graph)	C1	
$V = E \times R_Y / (R_Y + R_X)$ $= 9 \times 25 / (25 + 450)$ or $I = E / R_{\text{Total}} = 9 / (25 + 450) = 1.89 \times 10^{-2} \text{ A}$ $V = IR = 1.89 \times 10^{-2} \times 25$ or $V_{(X)} = 9 \times 450 / (25 + 450) = 8.53 \text{ V}$ $V = 9 - 8.53$	C1	
$V = 0.47 \text{ V}$	A1	 V [3]

- (iii) The resistance of X is increased. The temperature of Y remains at 190°C .

By reference to the current in the circuit, state and explain the effect of this change, if any, on the potential difference **across Y**.

(resistance of X increases so) the <u>total</u> resistance increases	B1
current decreases	B1
potential difference (across thermistor / Y) decreases	B1

.....
.....
..... [3]

[Total: 9]

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5 (a) (i) State Kirchhoff's second law.

sum of electromotive force(s) = sum of potential difference(s) around a (closed) loop
or
 the (algebraic) sum of the p.d.(s) and e.m.f.(s) is zero around a (closed) loop

B1

..... [1]

(ii) State the conservation law that gives rise to Kirchhoff's second law.

(law of conservation of) energy

B1 [1]

(b) A circuit contains a cell of internal resistance r and two resistors of resistances R_1 and R_2 , as shown in Fig. 5.1.

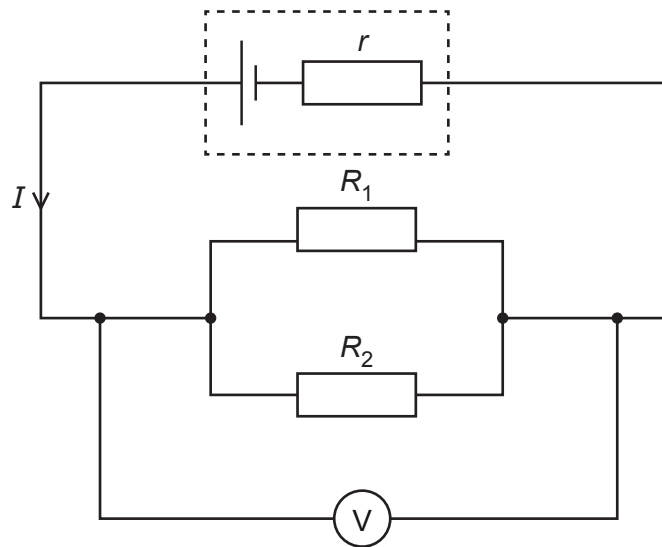


Fig. 5.1

The potential difference (p.d.) across the two resistors is V .

The current in the cell is I .

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(i) Use Kirchhoff's laws to show that the total resistance R_T of the external circuit is given by

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}.$$

(by Kirchhoff's first law) $I = I_1 + I_2$	B1
$V/R_T = V/R_1 + V/R_2$ therefore $1/R_T = 1/R_1 + 1/R_2$	B1

[2]

(ii) The electromotive force (e.m.f.) of the cell is 1.50 V.

When the values of R_1 and R_2 are $10\,\Omega$ and $15\,\Omega$ respectively, the p.d. measured by the voltmeter is 1.38 V.

Calculate the internal resistance r of the cell.

resistance of parallel combination $= (15 \times 10) / (15 + 10) \quad (= 6.0\,\Omega)$	C1
$r = (E - V) / I$	C1
$I = 1.38 / 6.0 = 0.23\text{ A}$ $r = (1.50 - 1.38) / 0.23$ $= 0.52\,\Omega$	A1
or	
(by potential divider principle) $r / R_T = Ir / V$	(C1)
$r / 6.0 = 0.12 / 1.38$ $r = 0.52\,\Omega$	(A1)
or	
(by potential divider equation) $V = E \times R_T / (R_T + r)$	(C1)
$1.38 = 1.5 \times 6.0 / (6.0 + r)$ $r = 0.52\,\Omega$	(A1)

Ω [3]

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(c) A third resistor is added in parallel with R_1 and R_2 in the circuit in Fig. 5.1.

State and explain the effect, if any, of this change on:

(i) the current in the cell

..... as the (total) resistance has decreased (and e.m.f. is unchanged)	M1
current will (in the cell) increase	A1

[2]

(ii) the p.d. measured by the voltmeter.

..... (as greater current means a) bigger drop in p.d. across the internal resistance	M1
p.d. (on voltmeter) will decrease	A1

[2]

[Total: 11]

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- 7 (a) Fig. 7.1 shows two resistors connected in series with a cell of electromotive force (e.m.f.) 1.50 V and internal resistance 0.28 Ω.

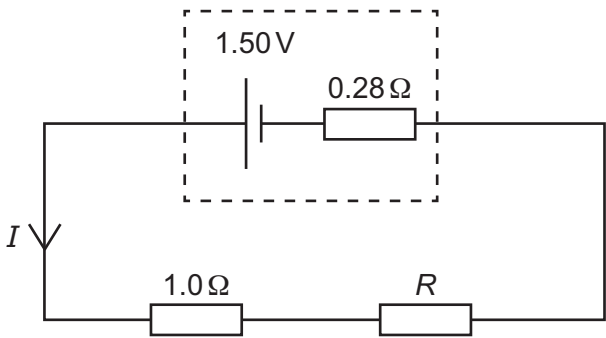


Fig. 7.1

One of the resistors has resistance 1.0 Ω. The other resistor has resistance R .
The terminal potential difference (p.d.) across the cell is 1.36 V.

- (i) Show that the current I in the circuit is 0.50 A.

(V =) 1.50 – 1.36 (= 0.14 V)	B1
current = V / R = 0.14 / 0.28 = 0.50 A	A1

[2]

- (ii) Calculate the combined resistance of the two resistors.

resistance = V / I	C1
= 1.36 / 0.50	A1
= 2.7 Ω	

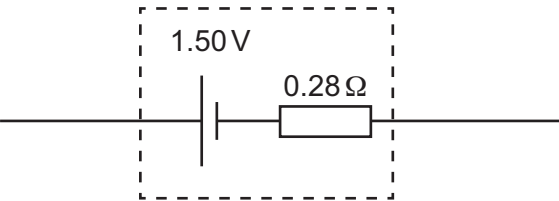
resistance = Ω [2]

- (iii) Use your answer in (a)(ii) to determine resistance R .

$R = 2.7 - 1.0$ = 1.7 Ω	A1
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$R =$ Ω [1]

- (b) The circuit in Fig. 7.1 is disconnected and the two resistors are reconnected to the cell, now in parallel with each other.
- (i) On Fig. 7.2, complete the circuit diagram to show this arrangement.



two resistors correctly shown in parallel with cell and no other components

B1

Fig. 7.2

[1]

- (ii) Explain, without calculation, whether the terminal p.d. across the cell is now less than, equal to or greater than 1.36 V.

(external) resistance is now smaller and (so) current (in cell) is greater / (external) resistance smaller fraction of total resistance / internal resistance larger fraction of total resistance	B1	
(greater p.d. across internal resistance so) terminal p.d. is less	B1	

[2]

[Total: 8]

5 (a) Define electric potential difference (p.d.).

energy transferred per unit charge (from electrical to other forms)	B1
.....	[1]

(b) A power supply, three resistors and a component X are connected in the circuit shown in Fig. 5.1.

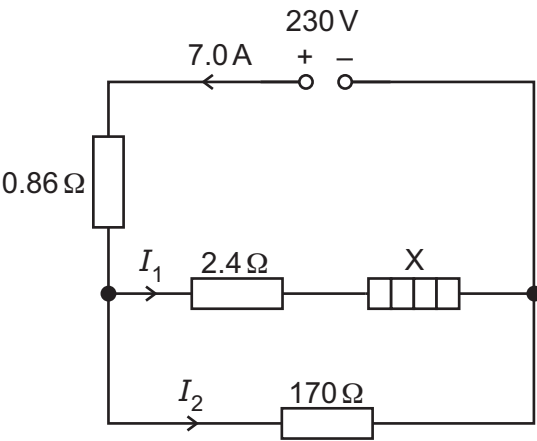


Fig. 5.1

The power supply has an electromotive force (e.m.f.) of 230V and negligible internal resistance. The current in the power supply is 7.0A.

(i) Identify component X.

heater	A1
.....	[1]

(ii) Show that the p.d. across the resistor of resistance 0.86Ω is 6.0V.

(V =) 7.0 × 0.86 = 6.0 (V)	A1
	[1]

(iii) Determine the current I_1 .

$I = (230 - 6.0) / 170$	C1
(= 1.3 A)	
$I_1 = 7.0 - 1.3$	A1
= 5.7 A	

$I_1 =$ A [2]

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(iv) Calculate the p.d. across component X.

$V = 230 - 6.0 - (5.7 \times 2.4)$	C1	[2]
$= 210 \text{ V}$	A1	
or		
$R = ((230 - 6.0) / 5.7) - 2.4$ $(= 36.9 \, \Omega)$	(C1)	
$V = 5.7 \times 36.9$ $= 210 \text{ V}$	(A1)	

(v) Calculate the power dissipated in component X.

$P = IV$ or $P = I^2R$ or $P = V^2 / R$	C1	A1
$P = 5.7 \times 210$		
or		
$P = 5.7^2 \times (210 / 5.7)$		
or		
$P = 210^2 / (210 / 5.7)$		
$P = 1200 \text{ W}$		

power = W [2]

(vi) The purpose of the circuit is to provide power to component X.

Determine the percentage efficiency of the circuit.

% efficiency = <u>useful</u> power out / (total) power in ($\times 100$)	C1	A1
$= 1200 / (230 \times 7.0) (\times 100)$		
$= 0.75 (\times 100)$		
$= 75\% \text{ or } 74\% \text{ (using 3 s.f. value from (v) gives } 74\%)$		

efficiency =% [2]

(vii) The resistor of resistance $170 \, \Omega$ is removed, leaving an open circuit in the lower branch of the circuit. There is no change to the resistance of component X.

State whether the current in the power supply increases, decreases or remains the same.

(current) decreases	A1	[1]
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[Total: 12]